

ISYS90045 Professional IS Consulting

Tutorial 9 Project Management

Tutorials led by Christina Parkes and
Leila Meratian Esfahani

Based on slides by Dr Leila Meratian Esfahani

Learning Objectives

- What are the component of Project Managements
- What are the different Project Management Methodologies and what are the differences?

What is a Project?

- An organised, temporary activity employing differently skilled people that is intended to introduce some justified, beneficial change to business operations (Axelos 2017).
- According to the Project Management Institute (PMI), a project is “a temporary endeavour undertaken to produce a unique product, service, or result ” (PMBOK® Guide , PMI, 2021, p. 245). Thus, a project should have :
 - definite starting and ending points (time)
 - budget (cost)
 - a clearly defined scope— or magnitude— of work to be done
 - specific performance requirements that must be achieved

Why Do Projects Need Managing?

Projects Management involve:

1. Scope: what does the project encompass? (and what is not)
2. Approach
3. Deliverables (products)
4. Time
5. Resources: Money and People
6. Risk
7. Governance

These variables are interdependent, i.e., change one variable and the other is affected. For example, expanding the scope may have an impact on quality of the deliverables, time, and cost.

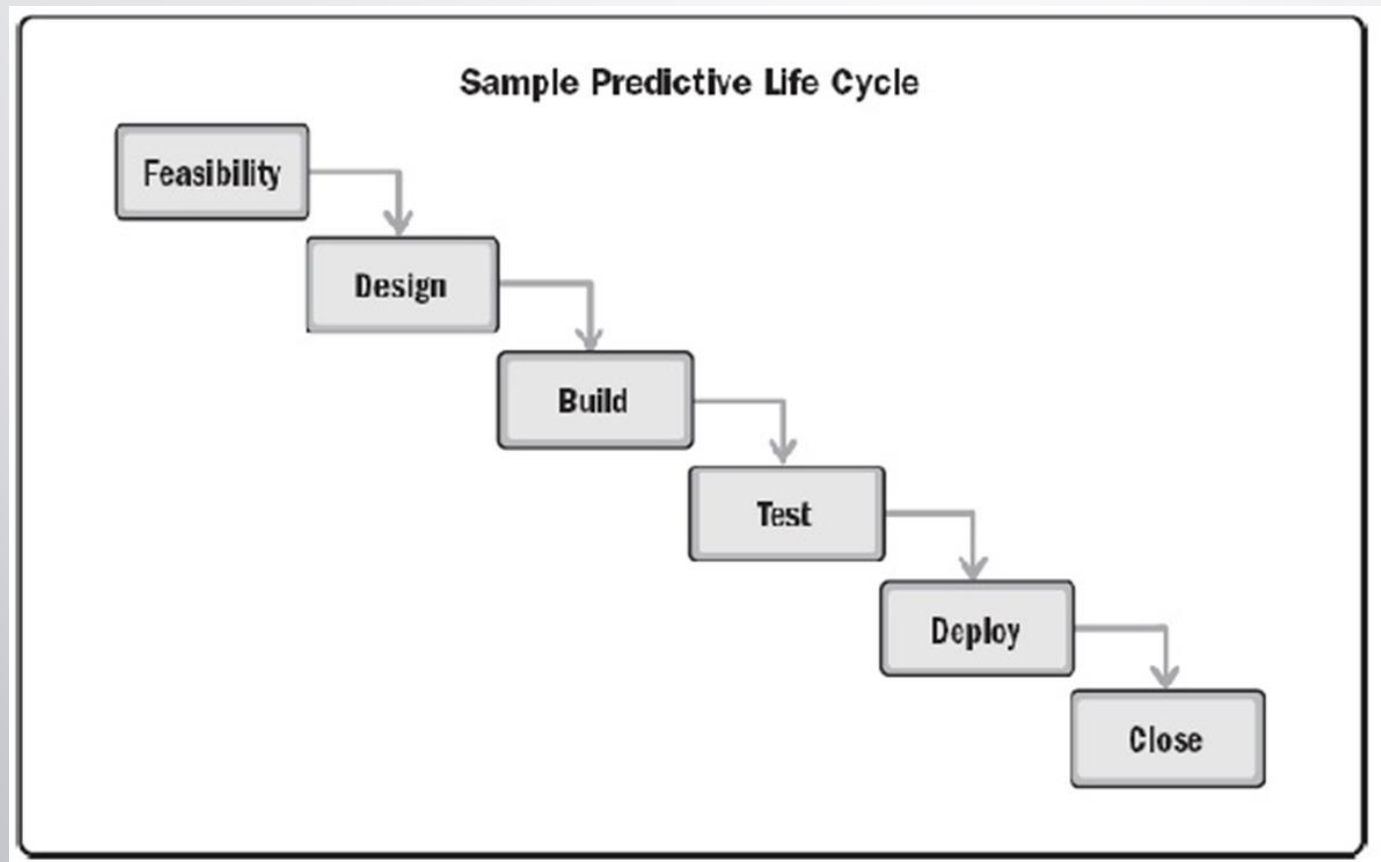
1. Scope

- Defining and controlling what is included in the project.
- Also, what is not included in the project
- Collect Requirements: the detailed requirements of the final product or service should assemble and itemized.

2. Approach (or Methodology)

- A **predictive approach (Waterfall)** is useful when the project and product requirements e.g., scope, schedule, cost, and resources would be determined up front, and changes (risk) would likely be minimal.
- In this approach, a life cycle where one phase finishes before the next one begins, so each phase is only performed once, and each phase focuses on a particular type of work.

Example of Waterfall Approach



Project Management Body of Knowledge (PMBOK)

Initiating

Define a new project or a new phase, identify stakeholder by obtaining authorization

Authorise the work
Project Charter

Planning

Develop an integrated project management plan to attain project objectives

Plan the work
Scope, Budget
Work breakdown schedule
Gantt Chart
Communication plan
Risk Management

Executing

Complete the work and satisfy the project objectives

Work the plan
Statuses and Tracking
Quality
Forecasts

Monitoring

Track and review the project progress and performance, manage change

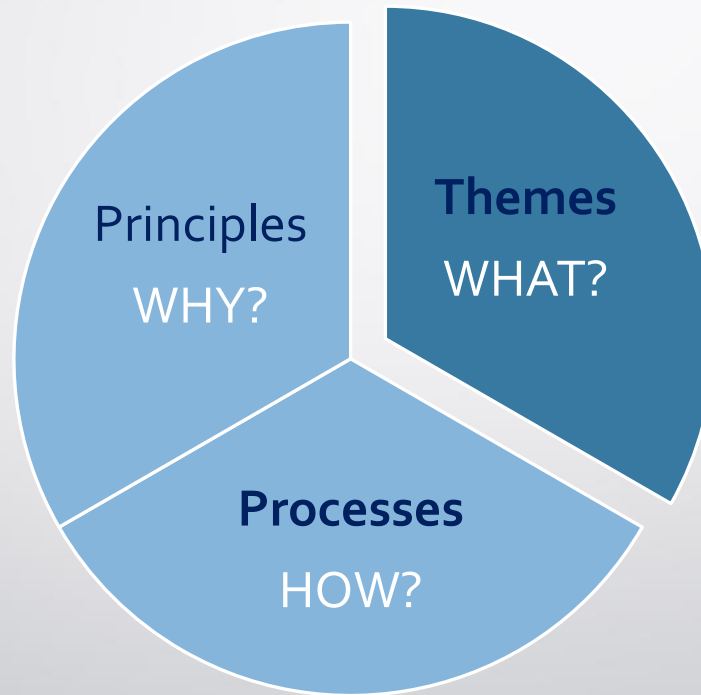
Control the plan
Quality Deliverable
Effort & cost tracking
Performance

Closing

Finalise all activities and formally close the project or phase

End the work
Post punch list
Reporting

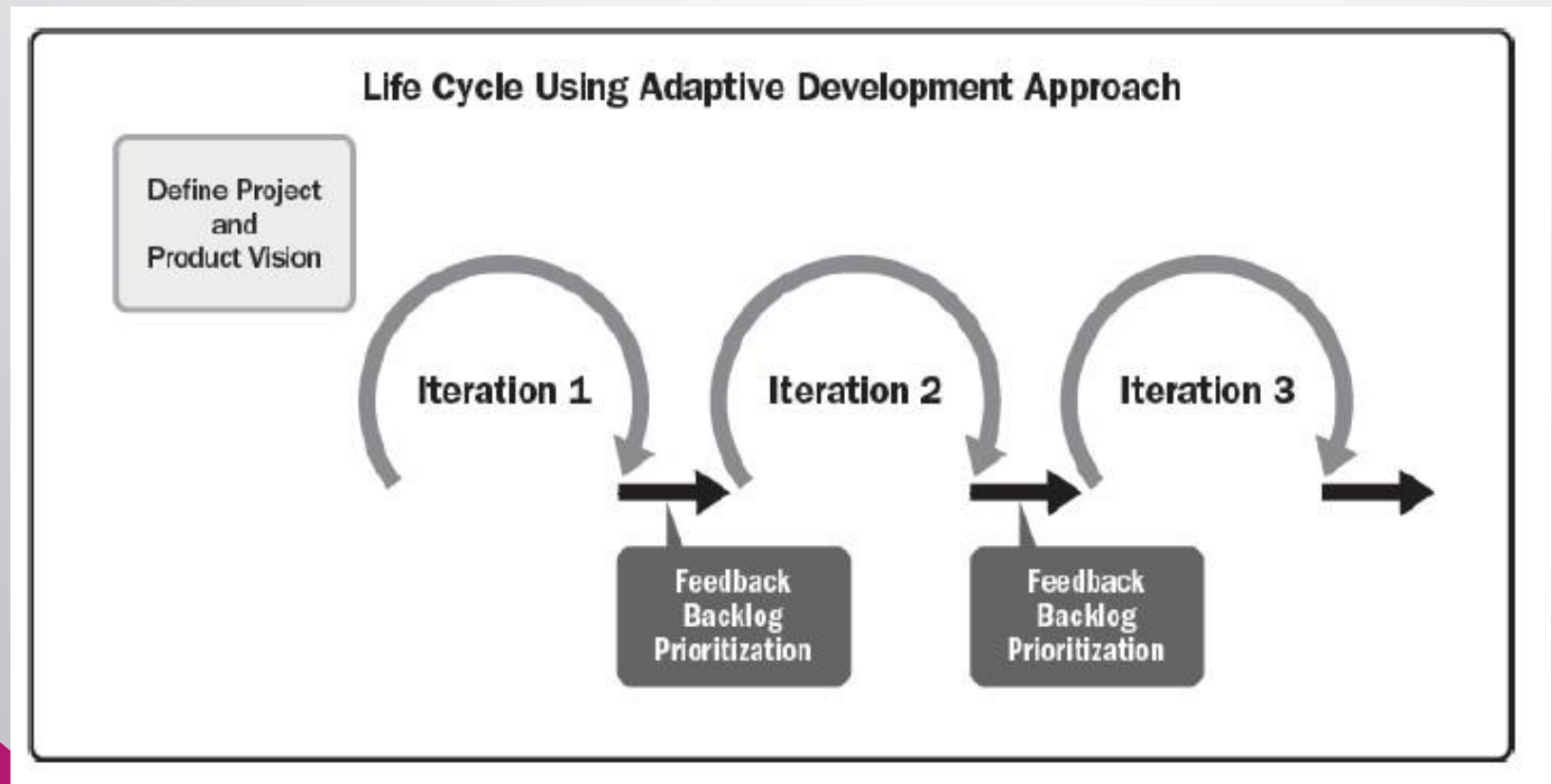
PRINCE2 (PRojects IN a Controlled Environment Version 2)



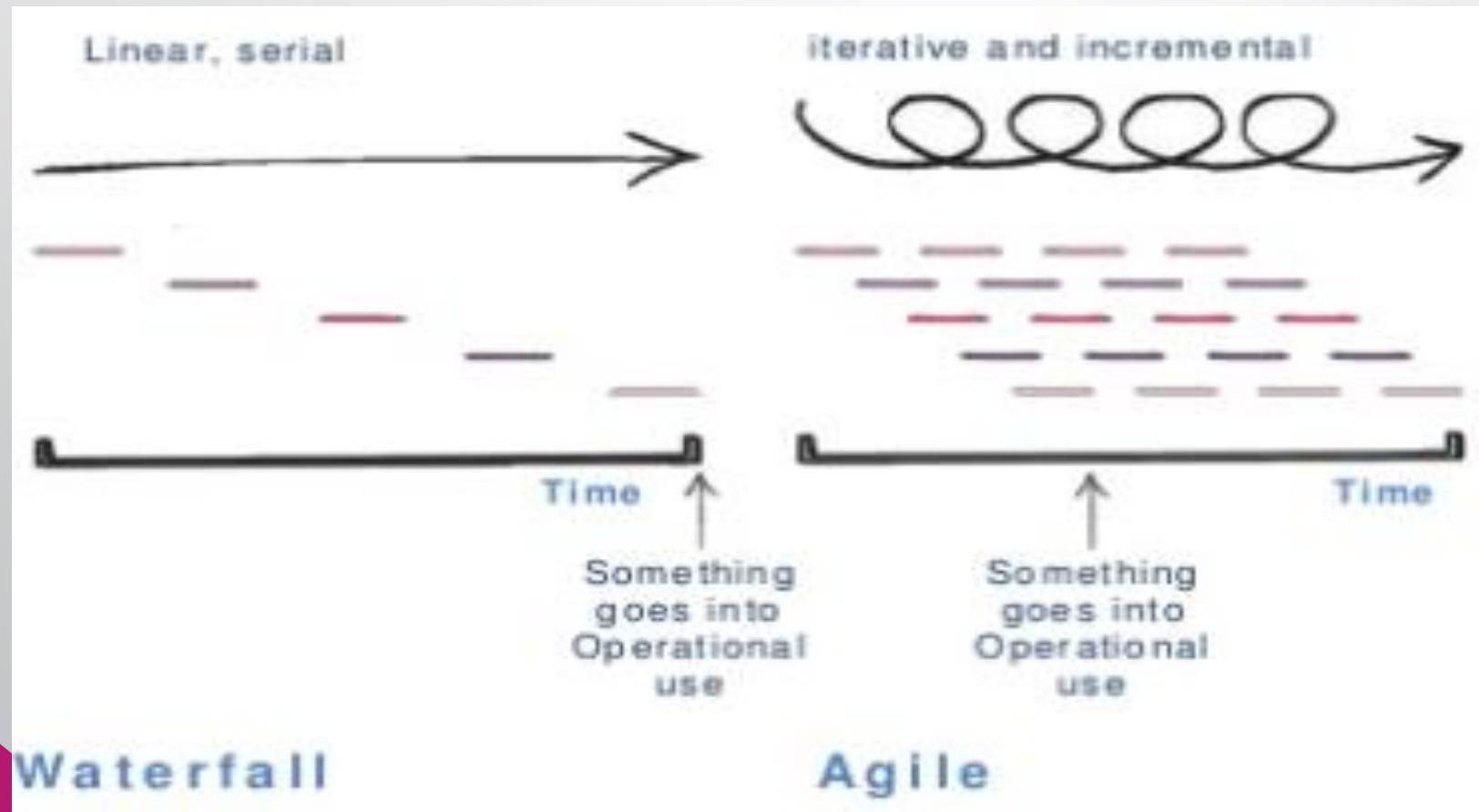
2. Approach (or Methodology)

- **Adaptive Approaches (Agile):** They are useful when requirements are subject to a high level of uncertainty and volatility and are likely to change throughout the project.
- A life cycle includes few iterations. At the end of each iteration (sometimes known as a sprint), the customer reviews a functional deliverable. At the review, the key stakeholders provide feedback, and the project team updates the project backlog of features and functions to prioritize for the next iteration.

Example of Agile Approach



Waterfall or Agile?



Agile (Iterative) Vs Waterfall (Linear Sequential Life Cycle)

- In the waterfall model, there is always a separate testing phase after a build phase
- However, in agile, development testing is usually done concurrently with, or at least in the same iteration as, programming
- Because testing is done in every iteration- which develops a small piece of the software-users can frequently use those new pieces of software and validate the value

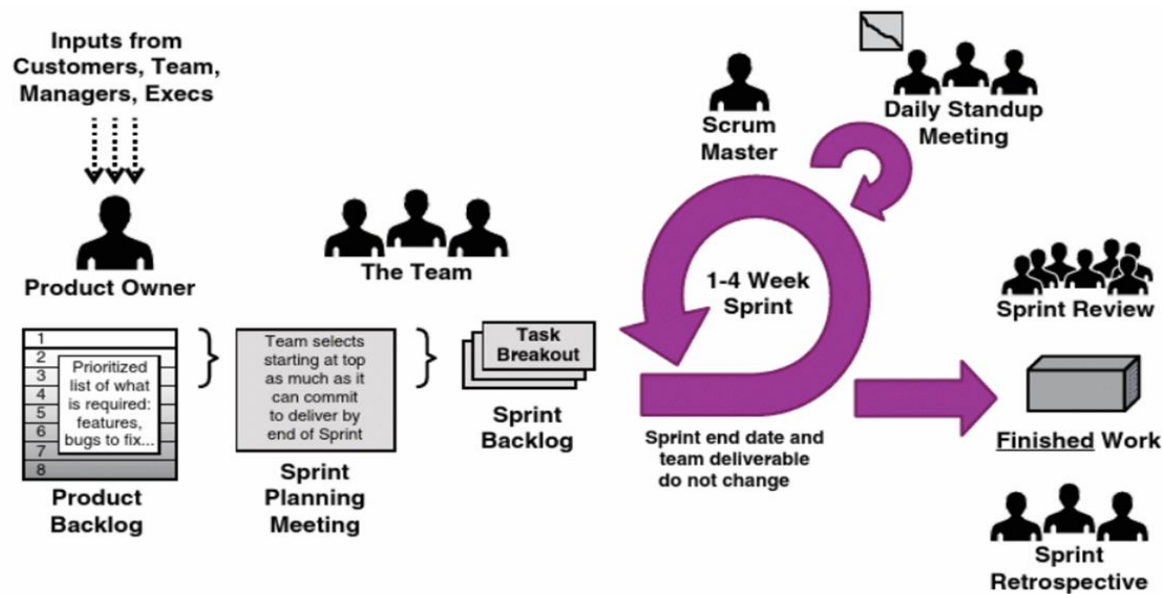
Why Agile:

- Provides opportunities to assess the direction of a project throughout the development
 - Achieved through regular cadences of work, known as sprints or iterations
 - At the end of a sprint teams must present a potentially shippable product increment
- Focusing on the repetition of delivered work cycles as well as the functional product they yield, agile methodology is described as iterative and incremental

Agile Manifesto

The Agile Manifesto		
Individuals and interactions	over	Processes and Tools
Working Product	over	Comprehensive Documentation
Customer Collaboration	over	Contract Negotiation
Responding to change	over	Following a plan

Agile Framework



Source: Cobb, Charles G.. The Project Manager's Guide to Mastering Agile : Principles and Practices for an Adaptive Approach, John Wiley & Sons, Incorporated, 2023.

3. Deliverables (products)

- Delivery means getting things done as per the project plan
- Deliverables are what the project delivers
- Project **deliverables**:
 - Internal: submitted to team member for internal review; e.g., initial design, project budget report, progress report
 - External: submitted to stakeholders, including clients and investors for client reviews; e.g., progress report, initial design, final design, final product.

4. Time

- Estimated time is a projected timeline based on how long similar tasks in past projects took to complete
- Divide the project into its component tasks, and then divide these component tasks into smaller tasks until a comprehensive list of things to do regarding the completion of the project is compiled
- Estimate the length of time each task will take – GANTT charts are useful here.
- Plan Schedule Management. The Schedule Management Plan contains information such as how the schedule will be created, who will be responsible for it, how aggressive it will be, and under what circumstances it will be changed

5. Resources: People

- Determination the skill-sets required of personnel to join the project team
- Determination of the number of personnel with a set of specific skill-sets, e.g. the number of programmers, or the number of UX designers
- Investigate the people already within the project or organisation that have these skills and their availability to work on the project at hand
- The remuneration of these personnel, i.e. the project budget needs to be able to accommodate their wages and salaries
- The level of motivation and enthusiasm amongst these people for the project, which is important for addressing the challenges and mitigating the risks associated with the project.

5. Resources: Money

- Costs associated with doing the work on the project, e.g. fees for services from third parties, salaries and wages of project personnel
- Costs associated with procurements and IT infrastructure, e.g. purchasing the necessary hardware and software for work on the project
- Estimate Costs. The cost of each task is estimated, taking into account the resources, labor, materials, equipment, and any other item of cost necessary to complete the task.

Next Week:

- 6. Risk
- 7. Governance

Tutorial:

- Project Management Methodologies
- https://www.youtube.com/watch?v=8UJ_qopwMDA

Tutorial:

- Think about a project and assume that you are assigned as the project manager. Then discuss the following questions in your team.
- What is the project?
- What manager should manage for this project?
- Who are the team?
- Who is the customer?
- What are the project deliverables?
- Discuss the 7 dimensions of a project.
- Outlining a Project Plan.

Group Activity:

Task: For each scenario, groups must decide: Would you use Agile, Waterfall?

Why is it a better fit? (consider risk, requirements clarity, stakeholder involvement, flexibility, compliance).

- **Case 1: Mobile Banking App:** A large commercial bank is planning to launch a new mobile app that allows customers to manage accounts, transfer money, and make digital payments. The features need to evolve based on customer feedback, and new regulatory requirements may arise during development. Time-to-market is critical because competitors are also preparing similar apps
- **Case 2: Airport Runway System:** A government aviation authority is commissioning the construction of a new runway and its associated control systems. Safety, compliance, and strict engineering standards are the highest priority, and requirements must be signed off before work begins. Any mistakes could have catastrophic consequences and would be extremely costly to fix.

- **Case 3: Online Learning Platform:** A university wants to create a new online platform to deliver courses, assessments, and collaboration tools to students worldwide. The design must adapt to student feedback, evolving pedagogy, and new technologies such as AI integration. Since the user base is diverse, flexibility and iteration are essential to meet varied needs.
- **Case 4: Hospital Patient Record System:** A public hospital is introducing an electronic patient record system to replace paper files and streamline medical workflows. The system must meet strict data security and privacy regulations, and integration with existing systems is non-negotiable. Doctors, nurses, and administrators will all rely on it, so accuracy and stability are critical.